Regenerative Engine PDR

Design Review, October 17, 2026



CAL POLY
SPACE SYSTEMS

Agenda

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Project Overview

- Regeneratively cooled N2O / ethanol engine, 300 lbf at sea level
- Fuel cools the chamber before injection (no film cooling, no ablatives, no single use components)
- Chamber printed as one piece in AlSi10Mg with integral cooling channels
- Goals: validate the thermal model, demonstrate repeated hot fires, and gather data
- With characterized engine, future engines can be built to support higher performance, flight vehicles, and other projects

Design Point

- Chamber pressure 200 psi, overall O/F 2.00
- Thrust 300 lbf sea level, delivered Isp 158 s
- Total flow 0.861 kg/s: 0.574 oxidizer, 0.287 fuel
- Running fuel rich trades 28 s of Isp for a flame 1,200 K cooler, along with more coolant
- Lower Pc reduces heat flux at cost of performance

Case	O/F	Flame temp	Delivered Isp
Design	2.00	1,894 K	157.8 s
Peak Isp	4.80	3,096 K	186.0 s

Requirements Flowdown

- 300 lbf sea level thrust drives throat area and total mass flow
- Cooling keeps the hot wall temperature safe for AlSi10Mg (Strength falls sharply after 300 °C)
- Bluegrass integration caps supply pressures: fuel 700 psi max, oxidizer self pressurized
- LPBF manufacturability sets a 1.0 mm floor on any printed geometry

Requirements & Verification (1/2)

#	Requirement	Rationale	Verification
1	Produce 300 lbf ±10% at sea level	Matches program goal, sizes stand load path	Test: Load cell during hot fire
2	Operate at Pc 200 psi ±10%	Sets throat size and all thermal loads	Test: Chamber pressure transducer during hot fire
3	Peak hot wall below 300 °C for ≥ 10 s	AlSi10Mg strength falls sharply above 300 °C	Test + analysis: Coolant and barrel temp during hot fire testing, analyze data
4	Bulk coolant subcooled at jacket outlet	Bulk boiling starves fuel flow and drives an O/F excursion	Test + analysis: Coolant temp and pressure during hot fire, analyze data
5	Jacket holds 700 psi with chamber at 0 psi, FoS ≥ 1.5	Worst case during fuel lead and cold flow	Test: Hydrostatic

Requirements & Verification (2/2)

#	Requirement	Rationale	Verification
6	No leaks at any seal at operating pressure	A leak across the ox/fuel boundary mixes propellants	Test: Hydrostatic and cold flow
7	All features printable at 1.0 mm minimum	Vendor process limit	Inspection: Sectioned sample
8	Jacket ΔP within the fuel supply budget	Fuel regulated to 700 psi maximum	Test: Cold flow with jacket inlet and outlet PTs
9	Integrates with Bluegrass mounts and plumbing	Test site compatibility	Inspection: Integration checked, demonstration in cold flow

End of Overview

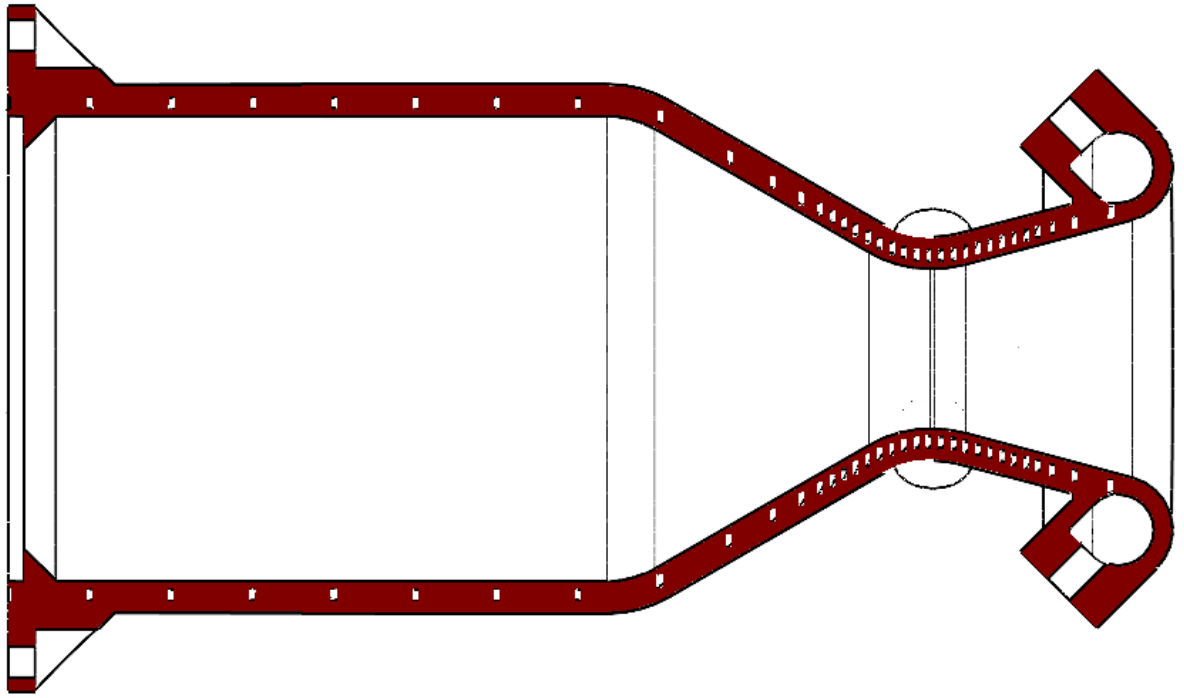
Questions?

Geometry & Cooling

Presented by Sloan Robbins

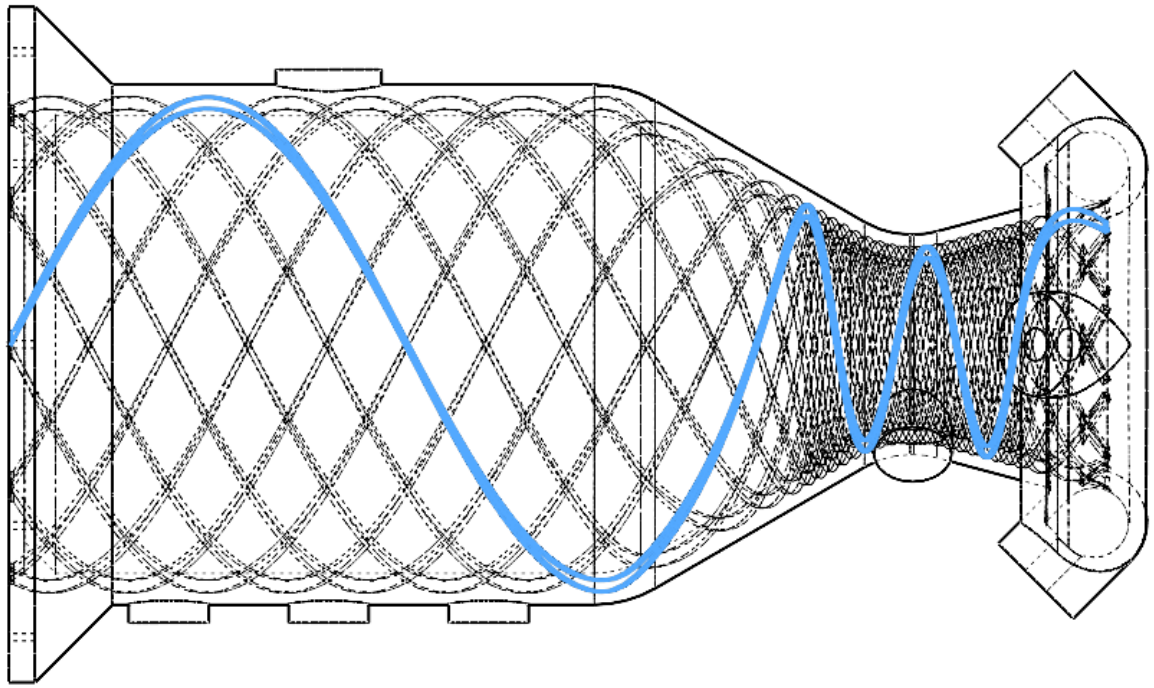
Chamber Geometry

- Chamber 89.0 mm (3.50 in) inner diameter
- Outer diameter 100 mm
- Throat 31.0 mm, exit 49.5 mm
- Expansion ratio 2.56 from optimum expansion: Pc / Pa = 200 / 14.7
- L* 1.2 m
- Contraction ratio 8.25, convergent 30°, divergent 15°
- Sized for sea level expansion using CEA outputs and MatLab script with standard NASA rocket equations



Cooling Channels

- 10 helical starts, constant 1.0 mm wide × 2.5 mm deep
- Hot wall 1.0 mm, land 1.0 mm at the throat, opening to 12 mm in the barrel to minimize path length (pressure drop)
- Helix angle is set by the land which is based on flux and coolant temp, effective channel density

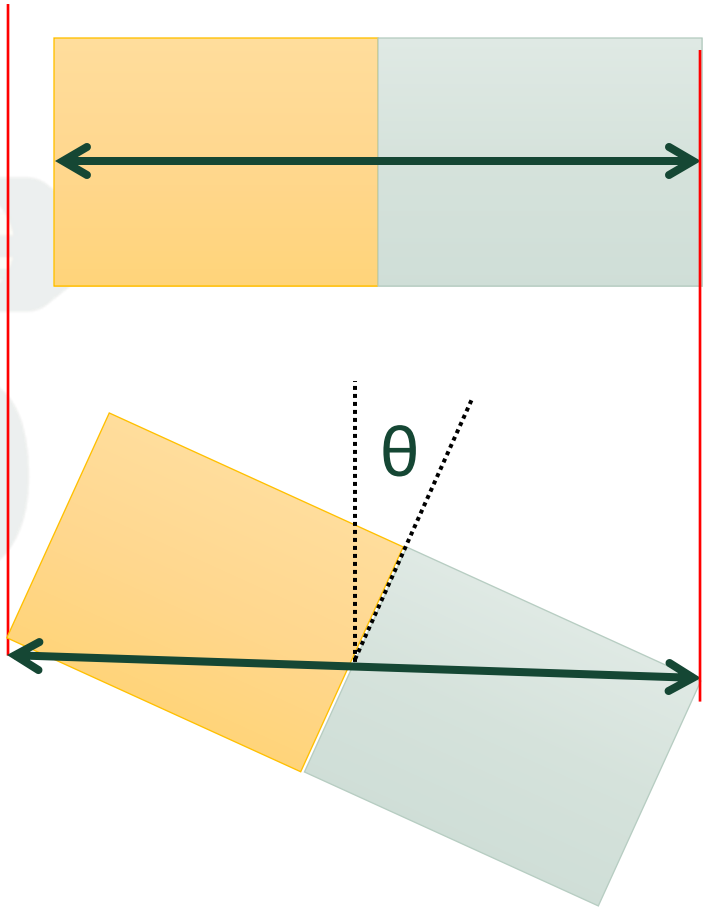


Channel Count and Size

- Lower limit of geometry is set at 1.0mm by manufacturing method
- Three channel variables yield two properties: width, depth, count -> ΔP, velocity
- Any increase in any variable decreases both pressure drop and velocity, therefore cooling
- Any decrease in any variable therefore increases performance, at the cost of higher feed pressures
- 10 × 1.0 × 2.5 is a balance of cooling performance with a “reasonable” pressure drop (~300psi)
- Supply pressure only acts as a lever for bulk coolant subcooling and has no impact on actual cooling performance.

Helical vs. Axial

- Land width fixes channel count
- Axial channels at the throat force about 49 starts, which means slower coolant and worse cooling
- The helix shrinks perpendicular pitch by cos θ, a ~5× gain in effective channel density
- Deep channels add sidewall without adding gas side footprint, which means the benefits of a larger channel without the downsides of a wide channel
- Cost is path length, nearly triple the axial, paid in feed pressure



Takeaway: Helical channels act to maximize channel density independent of actual channel size

Thermal Model

- Looped in MatLab from coolant inlet to injector, for entire length of engine.
- Gas side: Bartz with sigma correction, NASA CEA gas properties
- Coolant side: Gnielinski for Nusselt with Sieder-Tate viscosity correction, Haaland to calculate Darcy-Weisbach friction factor for roughness (assumed Ra of 6µm from vendor specs, converted to sand grain using standard industry model of $\varepsilon = 4 * Ra$, which is conservative)
- Lands treated as straight fins in heat transfer
- Solved for steady state heat transfer for cold wall and hot wall
- Hot Wall Flux -> Bulk Coolant Temp -> Cold Wall Temp -> Hot Wall Temp

Thermal Model Equations

Convective Heat Transfer Coefficient for Internal Fluid Flow

$$h = \frac{Nu \cdot k}{D_h}$$

Haaland Equation for Darcy-Weisbach Friction

$$\frac{1}{\sqrt{f}} = -1.8 log\left[\left(\frac{\varepsilon/D}{3.7}\right)^{1.11} + \frac{6.9}{Re}\right]$$

Gnielinski Correlation for Nusselt

$$Nu_{Dh} = \frac{(f/8)(Re_{Dh} - 1000)Pr}{1 + 12.7(f/8)^{1/2}(Pr^{2/3} - 1)}$$

Sieder-Tate Correlation for Nusselt (Viscosity Correction)

$$Nu_{Dh} = 0.027\, Re_{Dh}^{4/5} Pr^{1/3} \left(\frac{\mu}{\mu_s}\right)^{0.14}$$

Steady State Heat Transfer

Derived relationships between coolant temperature versus cold wall and cold wall versus hot wall

Steady State Heat Transfer

Cold Wall Coolant

$$Q_{IN} = Q_{OUT}$$

$$Q_{IN} = q''_{IN} \cdot A_{IN}$$

$$Q_{OUT} = h \cdot A_{OUT} \cdot \Delta T$$

$$q''_{IN} \cdot A_{IN} = h \cdot A_{OUT} \cdot \Delta T$$

$$\boxed{\Delta T = \frac{q''_{IN} \cdot A_{IN}}{h \cdot A_{OUT}}}$$

Hot to Cold Wall

$$Q = -k \cdot A \cdot \frac{dT}{dr}$$

$$Q = -k(2\pi r L)\frac{dT}{dr}$$

$$\frac{Q}{-2\pi k L} \cdot \frac{dr}{r} = \frac{dT}{1}$$

$$\frac{Q}{2\pi k L} \cdot \ln\left(\frac{R_2}{R_1}\right) = \Delta T$$

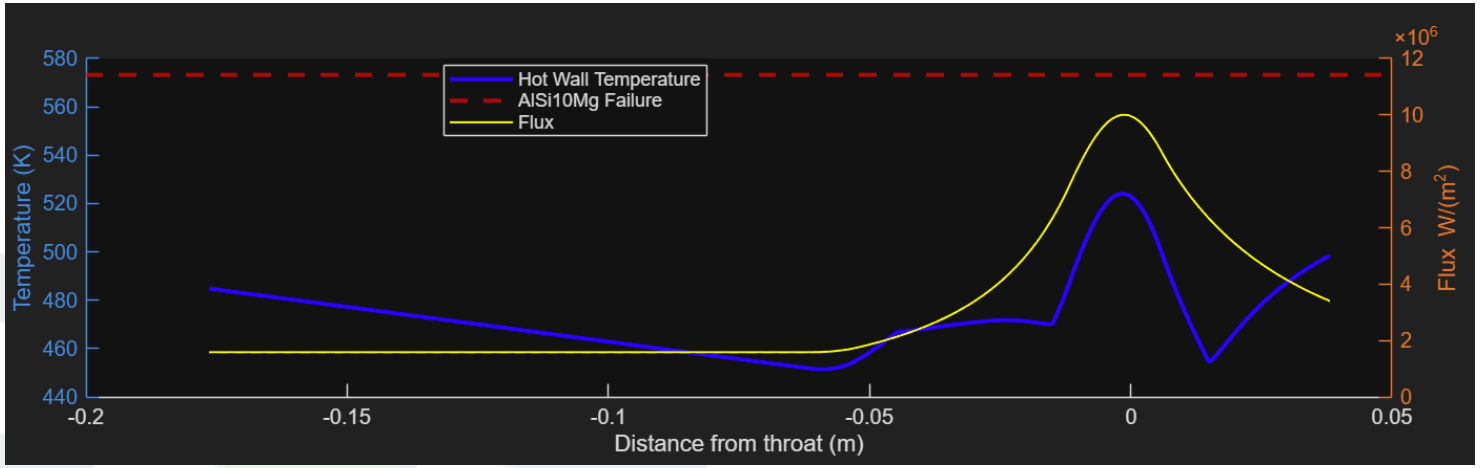
$$q''_{IN} = \frac{Q}{A_1}$$

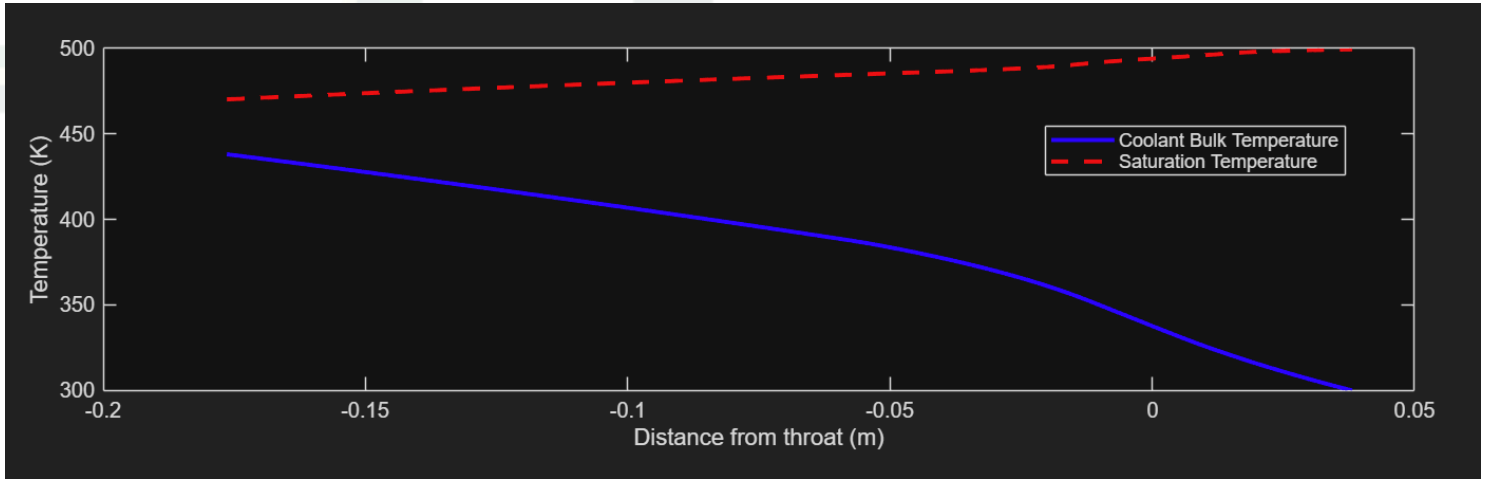
$$Q = q''_{IN} \cdot (2\pi R_1 L)$$

$$\boxed{\Delta T = \frac{q''_{IN} R_1}{k} \ln\left(\frac{R_2}{R_1}\right)}$$

Thermal Results

Heat load	124.5 kW
Coolant outlet	438.08 K
Coolant saturation	499.45 K
Peak cold wall	474.69 K
Peak hot wall	524.01 K
Friction pressure drop	292.2 psi
Channel length	612.5 mm





Takeaways:

- Bulk coolant is ~60K subcooled
- Hot wall is ~50K below “failure” temperature
- Pressure drop <300psi

Optimism vs. Pessimism

Optimistic: reality likely better	Pessimistic: reality could be worse
Bartz calculated at 1.0, small engines typically measure 0.6–0.8, which drastically drops flux [Smith; Lv et al.]	Without polishing inner bore, hot gas heat transfer will increase beyond what Bartz predicts for hydraulically smooth walls
Combustion assumed complete at the injector face, so barrel load is overstated	Injector streaks of oxidizer could put lean, hot mixture at the wall locally
No credit taken for wall thermal mass	Recirculation at the face to wall corner can raise local flux
Rougher than assumed channels cool better	Smoother channels cool worse

End of Geometry & Cooling

Questions?

Stresses & FOS

Presented by Ellen Poulsen

Channel Stress and Hoop Stress

- Thermal stress is more significant concern than mechanical stress, check worst case mechanical stress during fuel leads and cold flows (700psi in jacket, chamber at 0psi). Use hot yield temp for conservative calcs.
- Pressure stresses are small because each channel spans only 1.0 mm

CHANNELS:

σ_{max} OF FLAT RECTANGULAR PLATE

P = PRESSURE = 4.83 MPa
b = UNSUPPORTED WIDTH = 1.0mm
h = THICKNESS OF PLATE = 1.0mm

$$\sigma_{max} = \frac{3}{4} p \left(\frac{b}{h}\right)^2$$

$$\sigma_{max} = 3.40 \text{ MPa}$$

$$FOS = \frac{110}{3.40} = 32.4$$

HOOP STRESS:

$$\sigma_0 = \frac{P \cdot r}{t}$$

CLOSEOUT LOOP

	CYLINDER	THROAT
P	2.07MPa	1.38 MPa
r	44.5mm	15.5mm
t	2mm	2mm
	46MPa	10.69MPa
FOS	3.0	13

Location	Stress	Allowable	FoS
Wall bending over 1.0 mm channel due to pressure	3.4 Mpa	110 MPa at 250 °C	32.4
Closeout hoop, barrel	46 MPa	140 MPa at 150 °C	3.0
Closeout hoop, throat	10.69 MPa	140 MPa at 150 °C	13

Thermal Stress

$$S_c = \frac{(p_{co} - p_g)R}{t} + \frac{Eaqt}{2(1-v)k} \qquad (4\text{-}31)$$

where
- S_c = Max. combined compressive stress, lb/in.^{2}
- q = Heat flux, Btu/in.^{2}-s
- R = Radius of the inner shell, in.
- t = Thickness of the inner shell, in.
- p_{co} = Coolant pressure, lb/in.^{2}
- p_g = Combustion-gas pressure, lb/in.^{2}
- E = Modulus of elasticity of inner-shell material, lb/in.^{2}
- a = Thermal expansion coefficient of inner-shell material, in./in.-°F
- k = Thermal conductivity of inner-shell material, Btu/in.^{2}-s-°F/in
- v = Poisson's ratio of inner-shell material

"The maximum stress occurs at the inner-wall surface of the inner shell and can be calculated from this equation" (Huzel and Huang).

*Substitute with Fourier's Law of One-Dimensional Heat Conduction (use delta T = T_hotwall – T_coldwall

$q = k\Delta T / t$

$\therefore \; S_c = \frac{(P_{co} - P_g)R}{t} + \frac{E\alpha\Delta T}{2(1-\nu)}$

$\Delta T = 50.15\,K$
$R = 15.5\,mm$
$t = 1.00\,mm$
$P_{co} = 4.83\,MPa$
$P_g = 0.778\,MPa$
$E = 62\,GPa$
$\alpha = 22 \times 10^{-6}\,K^{-1}$
$\nu = 0.33$

$\Rightarrow S_c = 113.85\,MPa$

$\sigma_y = 110\,MPa$

$FOS = \frac{\sigma_y}{S_c} = \frac{110}{113.85} \approx 0.97$

- At maximum stress conditions and at 1mm wall thickness, FOS < 1
- Local yielding alone does not prove immediate elastic failure, should be paired with consideration of cycle fatigue determine how plastic deformation stabilizes and effects material over time

Life Cycle Fatigue

- Low Cycle Fatigue is a common point of failure from regenerative engines
- Complex analysis exists but often require datasets or more complex material properties from testing (such as exponents in Morrow Design Rule, only plastic deformation considered in Coffin Manson equation, elastic only in Basquin's equation).
- The Manson Universal Slopes Equation is a version of the Morrow Design rule with unknown exponents set as an average across metals.

$$\frac{\Delta\varepsilon_t}{2} = \frac{\Delta\varepsilon_e}{2} + \frac{\Delta\varepsilon_p}{2} = \frac{\sigma'_f}{E}(2N_f)^b + \varepsilon'_f(2N_f)^c$$

Example of Morrow Design Rule (elastic + plastic)

Life Cycle Fatigue

- Manson & Halford Research: how to accurately estimate high temperature low cycle fatigue, an application of linear damage analysis of fatigue and stress rupture
- 10% rule: life lower bound for thermal applications can be safely approximated as 10% of cycles found for non-thermal applications (accounting for thermal and creep)
- Therefore, given our low overall number of tests (and lower number of hot fires) results indicating a high FOS for cyclical loading can be used among other considerations to help validate safety of engine.

APPLICATION OF A METHOD OF ESTIMATING HIGH TEMPERATURE

LOW CYCLE FATIGUE BEHAVIOR OF MATERIALS

by G. R. Halford and S. S. Manson

In most cases, the lower bound life is given by 10 percent N_f and can be determined directly from the universal slopes equation proposed by one of the authors (2)

$$\Delta\epsilon_t = \frac{3.5\,\sigma_u}{E} N_f^{-0.12} + D^{0.6} N_f^{-0.60} \qquad (1)$$

where

$\Delta\epsilon_t$ = Total strain range
N_f = Cycles to failure
σ_u = Ultimate tensile strength
E = Modulus of elasticity
D = True tensile ductility [determined from the percent reduction of area, R.A., by $\ln\left(100/(100 - \text{R.A.})\right)$]

Life Cycle Fatigue Calcs

Calculated at throat, where thermal stress will be the greatest. Uses material properties at peak hot wall temp.

MANSON-HALFORD UNIVERSAL SLOPE EQUATION

$$\Delta\varepsilon_t = 3.5\frac{\sigma_{UTS}}{E}N_f^{-0.12} + D^{0.6}N_f^{-0.6}$$

⇒ SOLVE FOR N_f (# OF CYCLES TO FAILURE)

↳ $\Delta\varepsilon_t$ = TOTAL MECHANICAL STRAIN RANGE OVER ONE COMPLETE CYCLE

* EQUATION APPLIED @ THROAT, PEAK HOT WALL TEMP OF 250°C = 523.15 K

$\sigma_{UTS} = 250\,MPa$

$E = 62\,GPa$

T_L = MEAN LINER TEMP $= \frac{\text{HOT WALL} + \text{COOL WALL}}{2} = \frac{523.15 + 473.15}{2} = 498.15\,K$

T_c = TEMP COOLANT $= 337\,K$

$\Delta T_{LC} = T_L - T_c = 161.15\,K$

$\alpha = 22\times10^{-6}\,K^{-1}$ (ASSUMED)

FINDING ΔT_{hot} AND $\Delta\varepsilon_t$

$\alpha\Delta T = \varepsilon$ (HUZEL & HUANG) OR $\sigma = E\alpha\Delta T$ (FROM HOOKE'S LAW $\sigma = E\varepsilon$)

$\varepsilon_{mech} = \varepsilon_{total} - \alpha(T - T_0) \Rightarrow \Delta\varepsilon_t = \varepsilon_{mech,max} - \varepsilon_{mech,min}$

Life Cycle Fatigue Calcs

Hot wall wants to expand but is limited by wall around it. With temperature gradient across, hot wall is forced into compression.

Break thermal strain into two parts:

ε_1 (inner layer vs. outer layer): assume bonded layers see strain together (balancing forces). Use temperature value for inner wall, sees greater strain in compression, therefore uses compressed value.

ε_2 (hot face vs. mean inner temperature): 523.15 K − 498.15 K = 25 K, with bending ignored.

ε_1:

$\sigma = E(\varepsilon - \alpha \Delta T)$ t = THICKNESS, O = OUTER, I = INNER

$\Rightarrow \sigma_I t_I + \sigma_o t_o = 0$

$\Rightarrow E(\varepsilon_I - \alpha \Delta T_I) t_I + E(\varepsilon_o - \alpha \Delta T_o) t_o = 0$, $\varepsilon_I = \varepsilon_o$ (DEFORM TOGETHER)

$\Rightarrow \varepsilon = \dfrac{\alpha(\Delta T_I t_I + \Delta T_o t_o)}{t_I + t_o}$,

$\Rightarrow \varepsilon_1 = \varepsilon_{(common)} - \alpha(T_I - T_o) = \dfrac{-\alpha\ t_o}{t_I + t_o}(T_I - T)$, $t_o : t = 2 : 1$

$\Rightarrow \varepsilon_{inner} = -\alpha \frac{2}{3} \Delta T_o = (-\alpha 107.4)$ ← USE FOR ε_1, INNER WALL FACES GREATER COMPRESSION FORCES

ε_2:

$\varepsilon_2 = \alpha \Delta T$ (IN COMPRESSION)

$\varepsilon_2 = -\alpha(T_c - T_I)$

$\varepsilon_2 = -\alpha(523.15 - 498.075)$ ← BENDING SUPPRESSED

$\varepsilon_2 = -\alpha(25.0)$

$\Rightarrow \varepsilon_1 + \varepsilon_2 = \varepsilon$

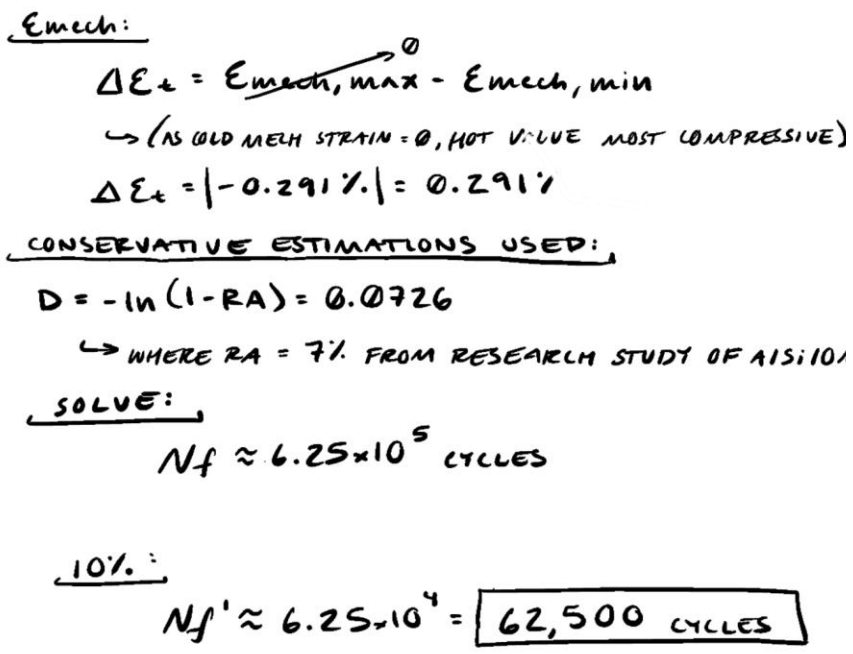
$\therefore \Delta T_1 + \Delta T_2 = \underline{\Delta T_{tot}}$

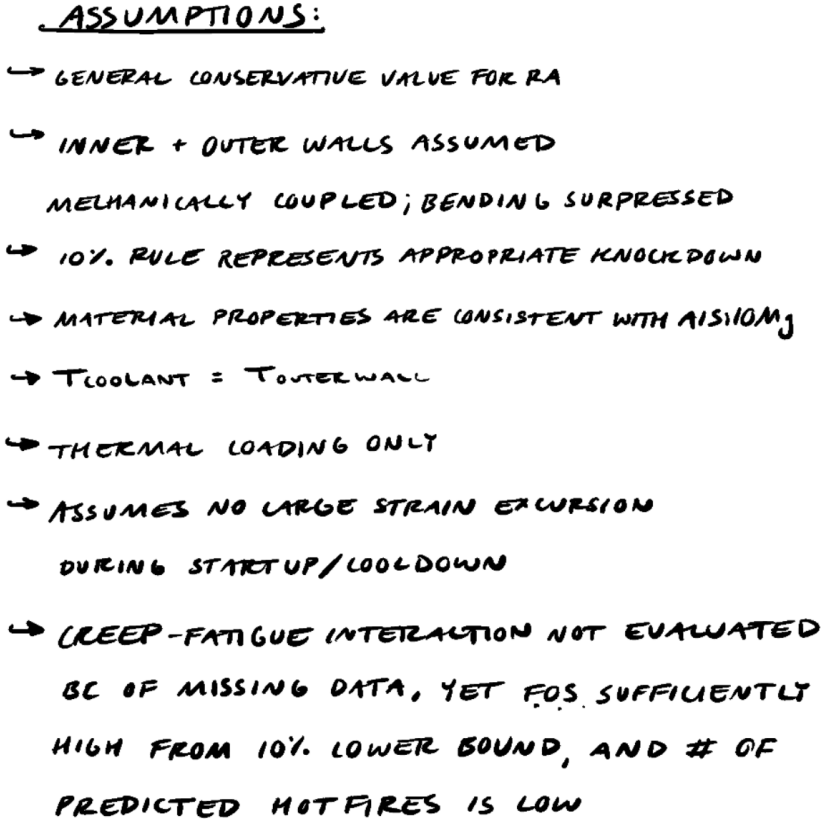
$\therefore 107.4 + 25.0 = \underline{132.4\ K}$

Life Cycle Fatigue Calcs

Strain range:
The wall starts cold at zero mechanical strain, and the hot condition is the most compressive.
The range is the magnitude of that swing, 0.291%.

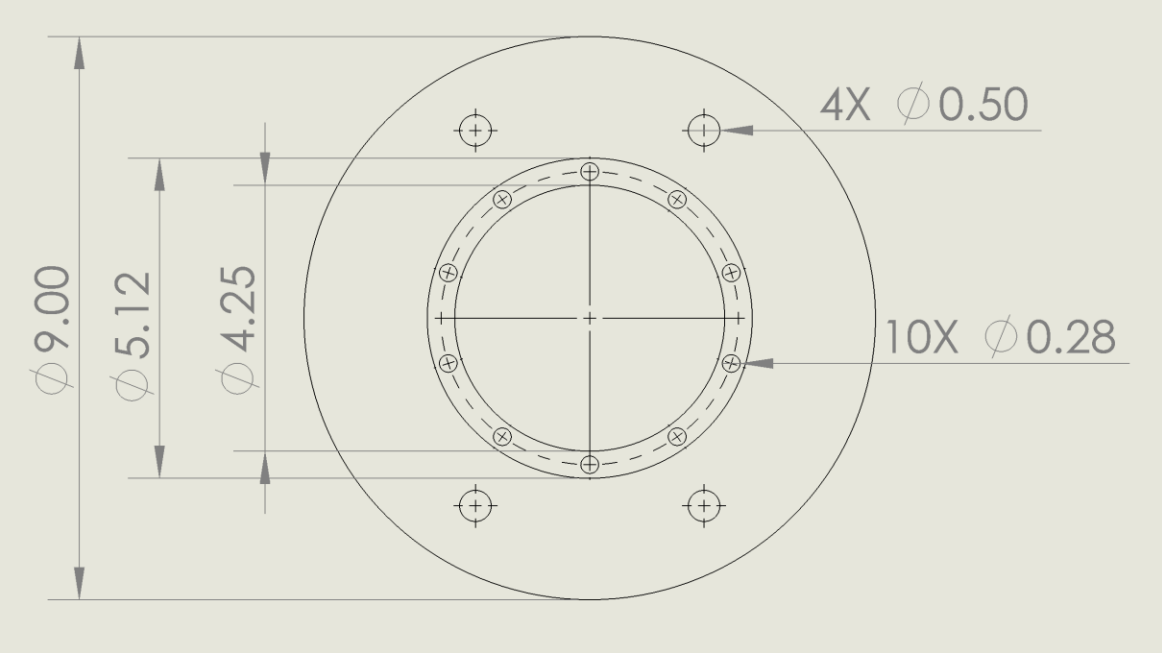
Solve: The equation gives Nf ≈ 6.25×10^5 cycles. 10% rule: Following Halford & Manson, the lower-bound life is 10% of that, so 62,500 cycles.

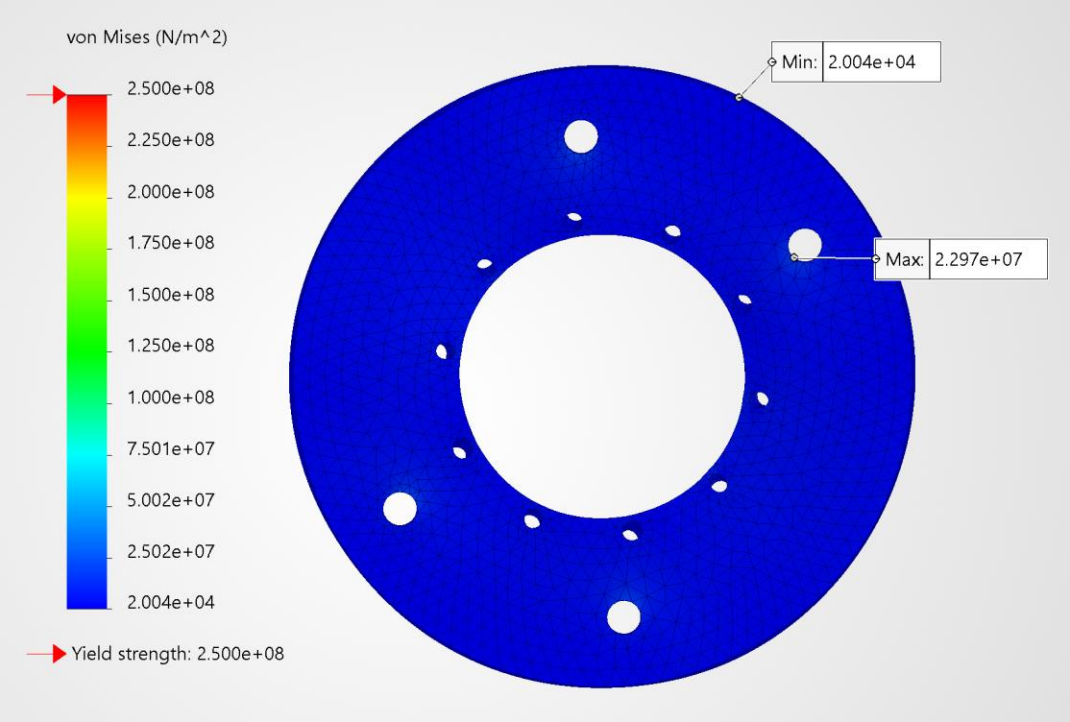




Integration Plate

- A36 Steel, 3/8"
- Holes for 10 ¼"-20 bolts
- Integration for Bluegrass test stand

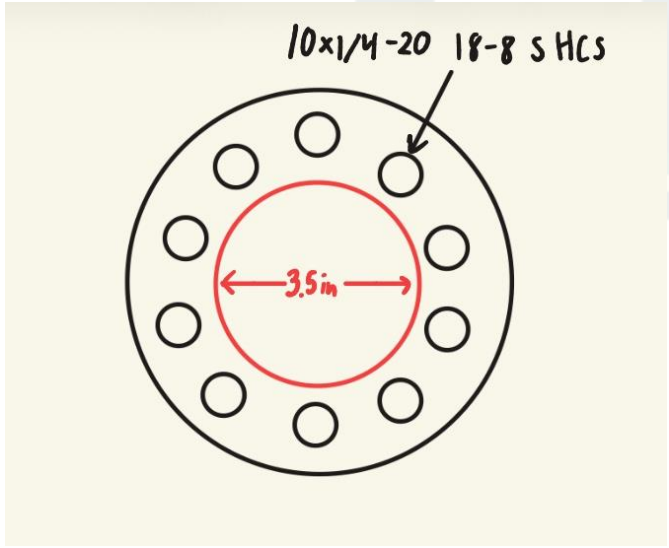




With 300lbf / 1334.47N applied on inner radius (where engine contacts plate), there is a FOS > 10

Bolted Joint

- 10x 1/4-20 stainless bolts with hex nuts, through bolted
- Per bolt yield: 2,230 lbf
- With 60% preload, conservative bolt FoS of 1.62



Bolt $\sigma_y = 70\text{-}80$ ksi

$A_{cs} = 0.0318\,in^2$

$F_{yield} = 2.23$ kips

Preload: $F_i \approx 0.6(2.23)$

$= 1.34$ kips

$F_i = 13.4$ kips for 10 bolts

Bore: $D = 3.5$ in, $A = 9.62\,in^2$

At 200psi P_c, $F_{sep} = 1924$ lbf

$C \approx 0.2$

$F_b = C\left(\frac{F_{sep}}{10}\right) = 38$ lbf

$F_{b,max} = 1.34 + 0.038 = 1.378$ kips

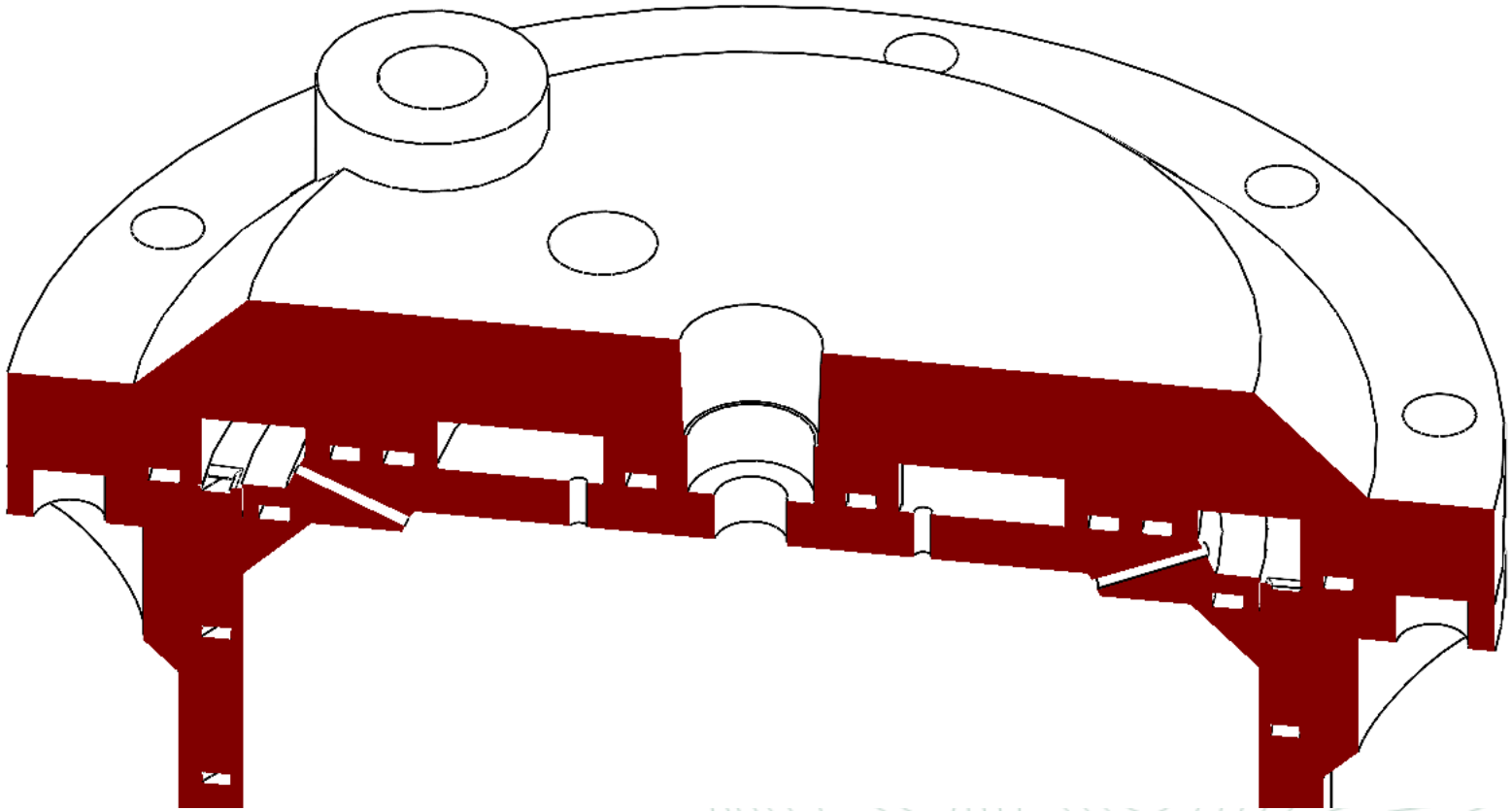
$FoS_b = \frac{2.23}{1.378} = 1.62$

End of Stresses & FOS

Questions?

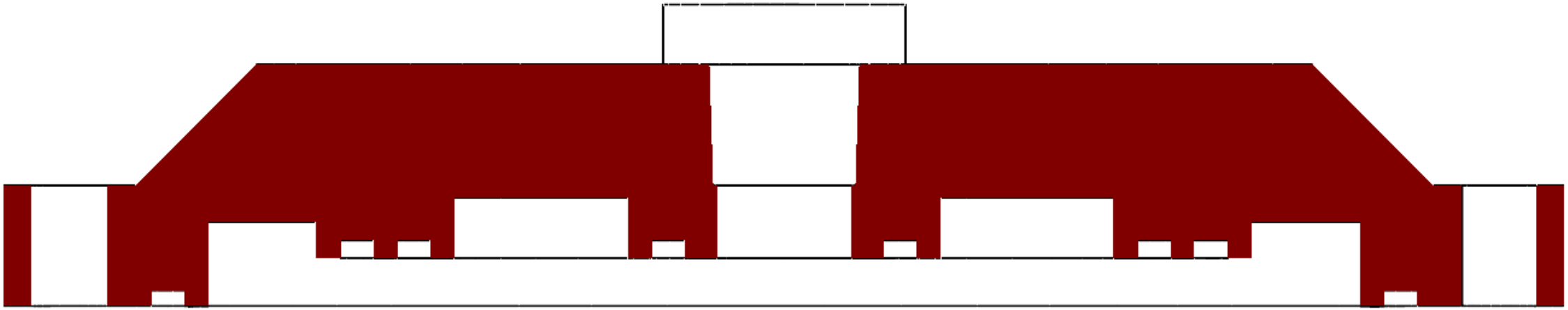
Injector & Plumbing

Presented by Wyatt Matthews



Sealing and O-rings

- AFLAS (TFE/P): copolymer O-ring compatible with ethanol and N2O, rated up to 204 °C against ~160 °C at the fuel outlet
- .07 in O-rings
- Face seals: 0.051 in x .106 in gland sizing gives 27% squeeze, 71% fill
- Double O-rings between ox/fuel and single O-rings between ignitor/ox and fuel/exterior



Injector Configuration

- 10 element pairs: 10 oxidizer orifices at 1.60 mm, 10 fuel orifices at 1.30 mm
- Fuel canted 65° into axial oxidizer, impinging ~8 mm off the face (about 5 ox diameters)
- $J = \rho_{ox} v_{ox}^2 / \rho_f v_f^2$, the ratio of the two jets' momentum flux
- At $J \approx 12$ the oxidizer jet dominates, so it shears and atomizes the fuel rather than colliding head on
  - (verified by cold flow)
- Machined in two pieces, so the plate can be re-drilled after cold flow or easily switched out for different designs



N2O Two-Phase Flow

- N2O orifice sizing calculated using Dyer two phase model
- Feed at 600 psi gives 400 psi across the ox injector, 200% of Pc
- Oxidizer orifices checked in cold flow with carbon dioxide to verify behavior

Dyer Two-Phase Model:

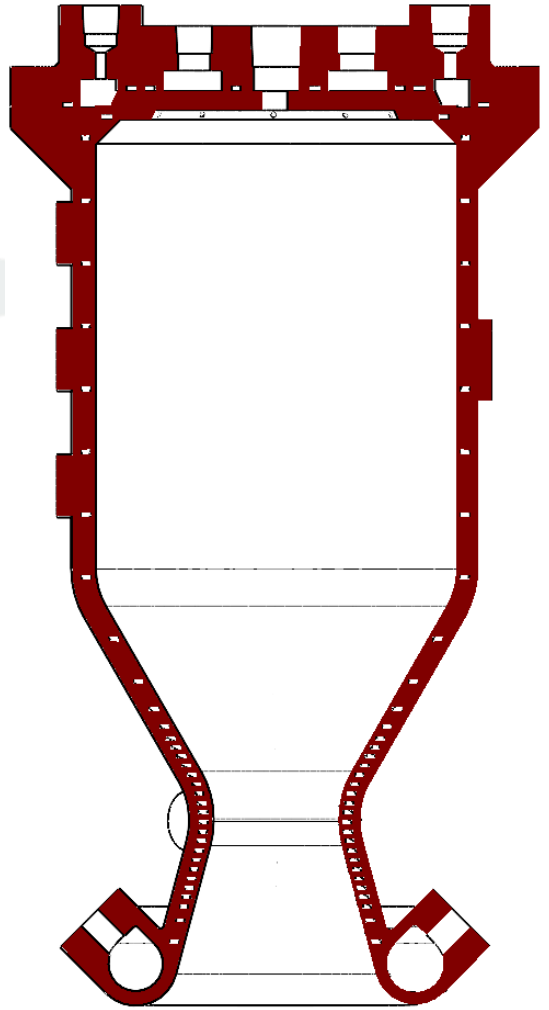
$$\dot{m}_{NHNE} = \left(1 - \frac{1}{1+\kappa}\right)\dot{m}_{SPI} + \frac{1}{1+\kappa}\dot{m}_{HEM}$$

Injector Manufacturing

- Carbide drills at the exact design sizes
- Carbide stays straight at L/D ≈ 10, HSS prone to wandering
- Each canted hole is spot faced normal to its axis, so the drill starts true
- Chamfered orifice inlets raise discharge coefficient
- Chamfered channel entries in the printed jacket cut entrance loss
- CNC available for top and bottom injector plate
  - Fuel orifices will be drilled

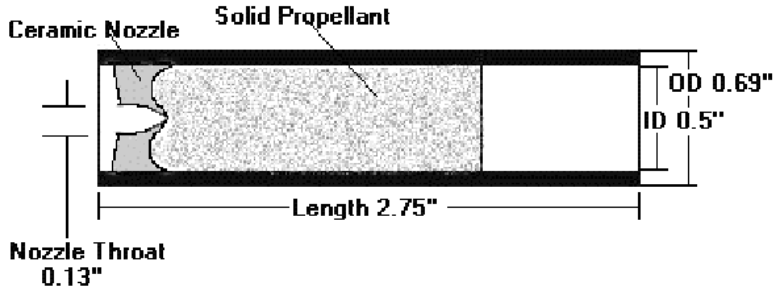
Feed System and Integration

- Oxidizer self-pressurized; injector feed assumed at 600 psi
- Fuel nitrogen regulated up to 700 psi
- Fuel enters at the nozzle end, leaves the jacket at the injector
- Budget: 200 psi Pc + 200 psi injector + 300 psi jacket = 700 psi, regulator set point fixed in cold flow
- Oxidizer enters the injector top
- Fuel stays inside the jacket and injector fuel manifold
- Both fuel and oxidizer use 2 × 1/8 NPT inlets to support uniform flow
- Adaptors will be manufactured to convert 1/4 NPT to 2 × 1/8 NPT



Ignitor

- Estes C6-0 engine fitted into a 1/4in NPT socket
- Nichrome wire ignition
- Burns for 1.6-1.9s
- Historically reliable design, used on all other CPSS rocket engines



End of Injector & Plumbing

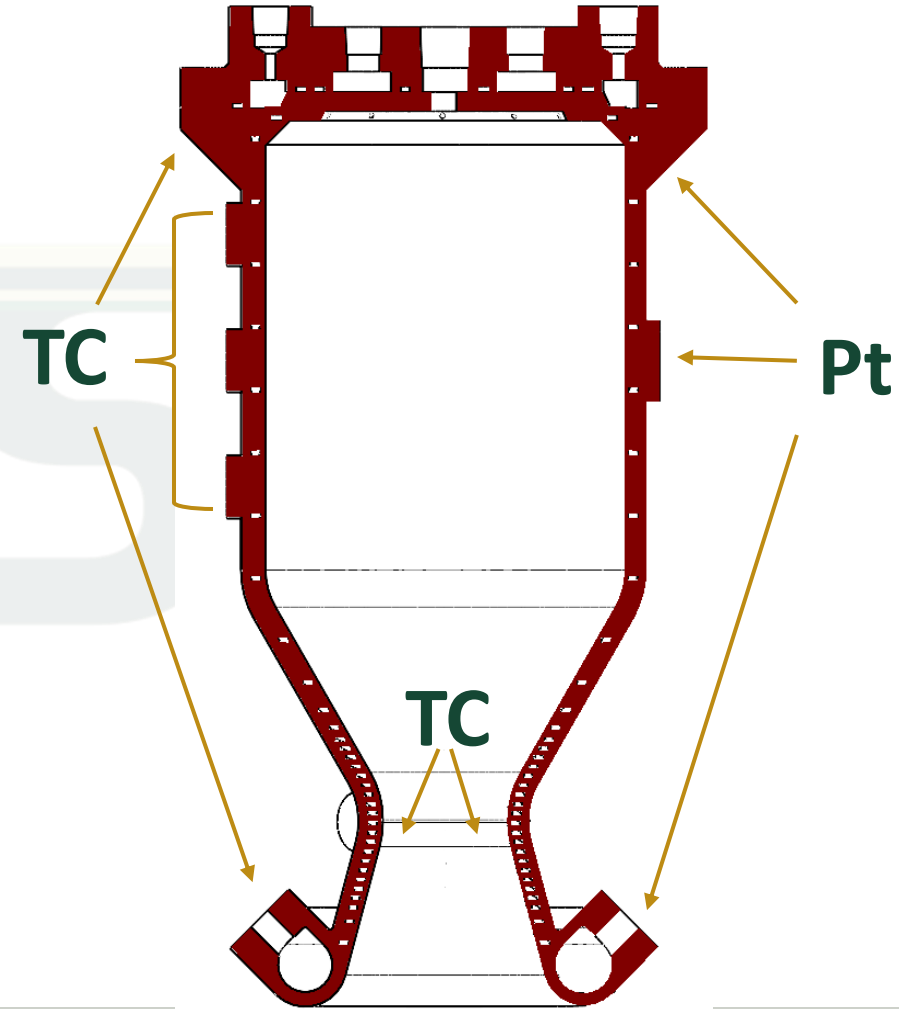
Questions?

Peripherals

Presented by Jeremiah

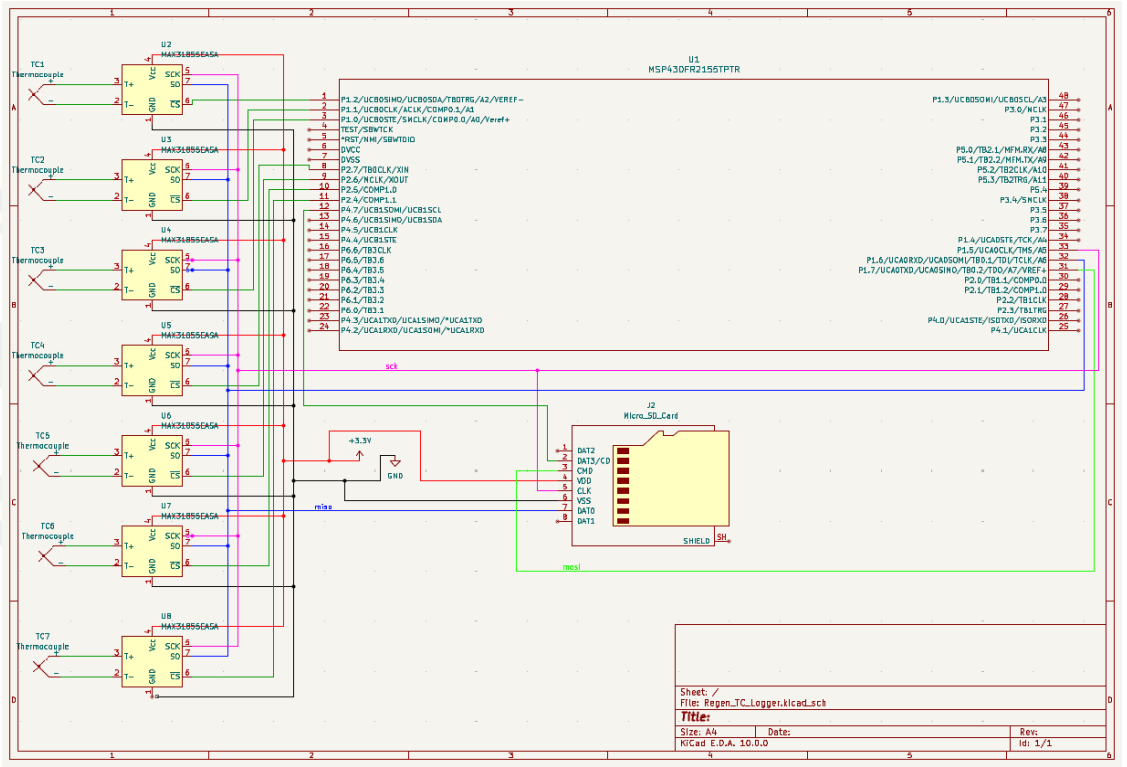
Instrumentation

- Pressure: Pc, fuel jacket inlet and outlet (jacket ΔP directly), both supplies
- Thrust: Load cells
- Coolant inlet/outlet thermocouples: Gives total heat load, which calibrates Bartz directly
- Three thermocouples: the model predicts a linear wall gradient there, and three points separate linear from curved
- Blind holes into lands, wall temperature extrapolated from hole depth
- Remaining 2 thermocouples on throat to extrapolate peak wall temp at worst station



TC Amplifier/Logger

- Save money by designing TC data acquisition system (~5x cheaper)
- 7 TCs, each connected to a MAX6675 amplifier
- Data logged to a micro-SD card
- Processing handled by TI MSP430 using SPI
- Using TI to get experience with the hardware



End of Peripherals

Questions?

Testing & Logistics

Presented by Jeremiah

Test Plan

Test	Goal	Duration	Success criteria
Cold flow 1: Injector + Ignitor	Ox vs. ignitor timing	Sequence	Verified timing and oxidizer flow
Hydrostatic	Verification	60 s	Jacket holds pressure and no leaks
Cold flow 2: Injector + Chamber	Jacket ΔP, roughness and regulator pressure	10 s steady	ΔP within budget and fuel flow
Cold flow 3: Chamber	Uniform channel flow + Cd	10 s steady	Verified channel discharge uniformity
Cold flow 4: Injector + Chamber	Ox vs. fuel timing	Sequence	Fuel lead confirmed
Cold flow 5: Injector + Chamber + Ignitor	Verification	5 s steady	Uniform flow and correct sequencing
Hot fire 1	Ignition and short burn	2–3 s	Wall and coolant at steady state, readings near model
Hot fires 2+	Duration ladder	5 → 10 → 20 s	Peak wall < 300 °C, coolant subcooled

Additive Manufacturing

- LPBF AlSi10Mg - CuCrZr equivalent quoted near $10k
- 1.0 mm minimum feature on wall, land and channel width
- Printed right side up, all surfaces sit >45° from horizontal and self support, only channel ceilings overhang
- Channels left as printed, since roughness helps cooling, throat bore polished
- Sample ordered to check channel dimensions, land fusion and preliminary roughness

Bill of Materials (1/2)

Item	Qty	Total	Justification
K-type thermocouple, TC-K6 (Auberins)	6	$57.30	Wall temperature in barrel lands and near the throat band
TC-K-1/8NPT probe (DwyerOmega)	2	$204.68	Coolant inlet and outlet temperature, gives total heat load
6061 disc, 5-1/4 in	2	$89.18	Injector body stock, one spare
6061 disc, 3-5/8 in	3	$44.07	Oxidizer plate stock, spares for other designs/remanufacture
Aflas O-rings, -045/-042/-040/-037/-017	12	$89.71	Ethanol and N2O compatible, rated 200 °C+
1/4-20 18-8 SHCS and hex nuts	1 pack	$49.45	Injector to chamber integration
Thermal paste	1	$14.99	Thermocouple contact in blind holes
Carbide drills	4	$68.72	Orifices at exact design size
Engine print, AlSi10Mg	1	~~$800.00~~	Sponsored by UnionFab
Total		**$618.10**	

Bill of Materials (2/2)

Item	Qty	Total	Justification
MSP430FR2355 Launchpad	1	$15.72	Read TC data and log to SD card
MAX6675 Amplifier	7	$83.3	Amplify TC signals
SD breakout board	1	$3.50	Integration
Total (electronics)		**$102.52**	

Total		**$720.62**	

End of Testing & Logistics

Questions?

Failures & Risks

Presented by Sloan

Failure Modes

Mode	Mechanism	Detection	Mitigation
Bulk coolant boiling	Heat load above coolant capacity	Outlet P drop, Pc drop	Supply pressure regulation up to 700 psi
Channel blockage	Powder left in a channel	Cold flow ΔP	Flow test print
Hot wall ratcheting	Cyclic thermal strain	Visual ridges between lands	Inspect between fires
Ox/fuel seal leak	O-ring failure	Cold flow testing	Double O-rings on every ox/fuel boundary

Open Risks

- Bulk coolant margin
- Injector face thermal load not analyzed
- Channel roughness and blockage unmeasured until cold flow
- Throat bore roughness

End of Failures & Risks

Questions?

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Final Questions?

No question is a bad question.
They will allow for us to critically think about our system and reason a proper solution to problems that might arise.











Thank you

Cal Poly Space Systems











